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Week 3 Intro

Lesson 1: Analysis of the Sample Mean

Lesson 2: General Framework of Statistical Estimation

Video: General Framework of Statistical Estimation

Video: Additional Probability II

Graded Assignment: Learning Check - General Framework of Statistical Estimation

Lesson 3: Central Limit Theorem

Weekly Assignment

Your grade: 100%

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Learning Check - General Framework of Statistical Estimation

Instructions

Review Learning Objectives

1. True or false: A statistic is typically a random variable.

1 / 1 point

True

False

Correct

Coach

Help me practice

Let's chat

Ready to review what you've learned before starting the assignment? I'm here to help.

Prior to obtaining data, there is uncertainty as to what value any particular statistic will take on, so a statistic is typically a random variable.

Assignment details

2. Suppose we take a sample of 100 students in the same class and try to estimate the average height of the class. Let's say this is Phase I. Next time, we take a sample of 100 students in the same class and try to estimate the average height of the class. Let's say this is Phase II. In which phase will the estimated average height have higher standard deviation?

1 / 1 point

Phase I

Phase II

Cannot be determined

Both will have the same variance

Correct

Coach

Help me practice

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The standard deviation is inversely proportional to the square root of the sample size so as the sample size increases, the standard deviation decreases.

3. Suppose the score distribution in a class follows a normal distribution with mean 85 and standard deviation of 5. Then, what score would you need to be in the 95th percentile of the class? (95% of all students have a score less than or equal to yours.)

1 / 1 point

76.8

95

85

93.2

Correct

Coach

Help me practice

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$$Z_{0.95} = \Phi^{-1}(1 - 0.05) = \Phi^{-1}(0.95) \text{ so } 0.95 = \int_{-\infty}^x \frac{1}{\sqrt{2\pi}} e^{-\frac{t^2}{2}} dt \text{ where } x \text{ is the score we are solving for. We can integrate this and solve for } x \text{ to get } x \approx 93.2$$

https://www.coursera.org/learn/statistics-for-data-science-essentials/assignment/17NAJ/learning-check-general-framework-of-statistical-estimation/view-feedback

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